

XML Practical 06 – Quick Reference

Simple table notes for viva and quick revision

1. Create XML Table

Code / Keyword	Simple Meaning
CREATE TABLE AdminDocs	Creates a new table called AdminDocs.
id NUMBER PRIMARY KEY	Creates a numeric id column and makes it the primary key.
xDoc XMLTYPE NOT NULL	Creates an XML column called xDoc. It cannot be empty.
XMLTYPE COLUMN xDoc STORE AS BASICFILE CLOB	Stores the XMLTYPE column as a BASICFILE CLOB.
/	Executes the command in Oracle tools.

2. Insert First XML Document

Code / Keyword	Simple Meaning
INSERT INTO AdminDocs VALUES (1, ...)	Inserts the first row into AdminDocs.
XMLTYPE(q'~ ... ~')	Treats the text inside as XML and converts it to XMLTYPE.
<catalog>	Root element of document 1.
<product dept="WMN">	Starts one product element. dept="WMN" is an attribute.
<number>557</number>	Stores the product number.
<name language="en">Fleece Pullover</name>	Stores the product name. language is an attribute.
<colorChoices>navy black</colorChoices>	Stores the color choices.
<desc>Our <i>favorite</i> shirt!</desc>	Shows nested XML. The i tag is inside desc.
</product>	Closes the product element.
</catalog>	Closes the root element.

3. Insert Second XML Document

Code / Keyword	Simple Meaning
INSERT INTO AdminDocs VALUES (2, ...)	Inserts the second row into AdminDocs.
<doc id="123">	Root element of document 2. id is an attribute.
<sections>	Groups all section elements together.
<section num="1"><title>XML Schema</title></section>	Creates one section with an attribute and a title child.
COMMIT;	Saves the inserted data permanently.

4. Display Stored XML

Code / Keyword	Simple Meaning
SELECT id	Displays the id column.

Code / Keyword	Simple Meaning
XMLSERIALIZE(CONTENT xDoc AS CLOB INDENT SIZE = 2)	Converts XMLTYPE data into readable text format.
FROM AdminDocs	Reads the rows from AdminDocs.

5. XPath Basics

Code / Keyword	Simple Meaning
/catalog/product	Go from the root catalog to product.
//product	Find product anywhere in the XML.
XMLQUERY('xpath' PASSING xDoc RETURNING CONTENT)	Runs the XPath on xDoc and returns the matching XML content.
*/product	Any root element, then product under that root.
*/product[@dept="WMN"]	Selects only products whose dept attribute is WMN.
*/child::product[attribute::dept="WMN"]	Same idea using axis notation.

6. XPath Filters and Position

Code / Keyword	Simple Meaning
//product[number > 500]	Returns whole product nodes where number is greater than 500.
//product/number[. > 500]	Returns only the number elements greater than 500.
/catalog/product[4]	Returns the 4th product under catalog.
//product[number > 500][@dept="ACC"]	Returns products where number is greater than 500 and dept is ACC.
//product[number > 500][1]	Returns the first product whose number is greater than 500.

7. XQuery Basics

Code / Keyword	Simple Meaning
XPath	Points to a location in XML.
XQuery	A full query language for XML.
for \$prod in //product	for means loop. \$prod is a variable over all product nodes.
let \$x := \$prod/number	Stores the product number in variable \$x.
where \$x > 500	Filters only values greater than 500.
return \$x	Returns the result.
data(\$x)	Extracts only the text value of the node.
if (\$x > 500) then <book>...</book> else <paper>...</paper>	Creates different XML output based on a condition.

8. XML DML Updates

Code / Keyword	Simple Meaning
UPDATE AdminDocs	Updates the table.

Code / Keyword	Simple Meaning
SET xDoc = insertXMLafter(...)	Inserts a new XML node into xDoc.
/doc/sections/section[@num="1"]	Finds the section whose num attribute is 1.
XMLTYPE('<section num="2"><title>Backg round</title></section>')	This is the new XML node to insert.
SET xDoc = deleteXML(...)	Deletes a node from the XML document.
/doc/sections/section[@num="2"]	Finds the inserted section with num = 2 and deletes it.

9. Most Common Viva Questions

Code / Keyword	Simple Meaning
What is XMLTYPE?	It is Oracle's data type for storing XML.
Why do we use XMLSERIALIZE?	To convert XML into readable text.
What is XPath?	It is used to select nodes from XML.
What is XQuery?	It is used to query XML and create XML output.
What does [@dept="WMN"] mean?	It filters nodes by attribute value.
What does data(\$x) do?	It gets only the text value of a node.
What does insertXMLafter do?	It inserts a new XML node after a selected node.
What does deleteXML do?	It removes a selected node from XML.